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PERSONALIZED GENERATIVE VIDEO SUMMARIZATION

Abstract

A method includes generating a multimedia content result using a streaming multimodal video generation model, including creating prompt text based on one or more user preferences, generating image frames based on the prompt text, compiling the image frames into the multimedia content result, and displaying the multimedia content result. The method can also include acquiring video data and audio data associated with a multimedia content item, performing a first scene detection process on the video data, performing a second scene detection process on the audio data, determining combined scene detection results based on the first scene detection results and the second scene detection results, inputting at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model, and generating the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION AND PRIORITY CLAIM [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/561,254 filed on Mar. 4, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to machine learning systems. More specifically, this disclosure relates to personalized generative video summarization.

BACKGROUND

[0003] Streaming television has become popular for users to consume linear television video content. However, users increasingly have less time to watch long-form video content, leading to the rise in short-form video content and services. There is, therefore, a need to provide users with a way to consume television content in a short-form format.

SUMMARY

[0004] This disclosure relates to personalized generative video summarization.

[0005] In one example, a method includes generating a multimedia content result using a streaming multimodal video generation model. Generating a multimedia content result using a streaming multimodal video generation model includes creating prompt text based on one or more user preferences, generating image frames based on the prompt text, compiling the image frames into the multimedia content result, and displaying the multimedia content result on a display.

[0006] In some examples, the method also includes acquiring video data and audio data associated with a multimedia content item, performing a first scene detection process on the video data to produce first scene detection results, performing a second scene detection process on the audio data to produce second scene detection results, determining combined scene detection results based on the first scene detection results and the second scene detection results, inputting at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model, and generating the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.

[0007] In another example, an electronic device includes at least one processing device configured to generate a multimedia content result using a streaming multimodal video generation model. The at least one processing device is further configured to create prompt text based on one or more user preferences. The at least one processing device is further configured to generate image frames based on the prompt text. The at least one processing device is further configured to compile the image frames into the multimedia content result. The at least one processing device is further configured to display the multimedia content result on a display.

[0008] In some examples, the at least one processing device is further configured to acquire video data and audio data associated with a multimedia content item, perform a first scene detection process on the video data to produce first scene detection results, perform a second scene detection process on the audio data to produce second scene detection results, determine combined scene detection results based on the first scene detection results and the second scene detection results, input at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model, and generate the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.

[0009] In another example, a non-transitory machine readable medium includes instructions that when

executed cause at least one processor of an electronic device to generate a multimedia content result using a streaming multimodal video generation model. The instructions when executed further cause the at least one processing device to create prompt text based on one or more user preferences, generate image frames based on the prompt text, compile the image frames into the multimedia content result, and display the multimedia content result on a display.

[0010] In some examples, the instructions when executed further cause the at least one processor of the electronic device to acquire video data and audio data associated with a multimedia content item, perform a first scene detection process on the video data to produce first scene detection results, perform a second scene detection process on the audio data to produce second scene detection results, determine combined scene detection results based on the first scene detection results and the second scene detection results, input at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model, generate the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.

[0011] Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

[0012] Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like.

[0013] Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

[0014] As used here, terms and phrases such as “have,” “may have,” “include,” or “may include” a feature (like a number, function, operation, or component such as a part) indicate the existence of the feature and do not exclude the existence of other features. Also, as used here, the phrases “A or B,” “at least one of A and/or B,” or “one or more of A and/or B” may include all possible combinations of A and B. For example, “A or B,” “at least one of A and B,” and “at least one of A or B” may indicate all of (1) including at least one A, (2) including at least one B, or (3) including at least one A and at least one B. Further, as used here, the terms “first” and “second” may modify various components regardless of importance and do not limit the components. These terms are only used to distinguish one component from another. For example, a first user device and a second user device may indicate different user devices from each other, regardless of the order or importance of the devices. A first component may be denoted a second component and vice versa without departing from the scope of this disclosure.

[0015] It will be understood that, when an element (such as a first element) is referred to as being (operatively or communicatively) “coupled with/to” or “connected with/to” another element (such as a second element), it can be coupled or connected with/to the other element directly or via a third element. In contrast, it will be understood that, when an element (such as a first element) is referred to as being “directly coupled with/to” or “directly connected with/to” another element (such as a second element), no other element (such as a third element) intervenes between the element and the other element.

[0016] As used here, the phrase “configured (or set) to” may be interchangeably used with the phrases “suitable for,” “having the capacity to,” “designed to,” “adapted to,” “made to,” or “capable of” depending on the circumstances. The phrase “configured (or set) to” does not essentially mean “specifically designed in hardware to.” Rather, the phrase “configured to” may mean that a device can perform an operation together with another device or parts. For example, the phrase “processor configured (or set) to perform A, B, and C” may mean a generic-purpose processor (such as a CPU or application processor) that may perform the operations by executing one or more software programs stored in a memory device or a dedicated processor (such as an embedded processor) for performing the operations.

[0017] The terms and phrases as used here are provided merely to describe some embodiments of this disclosure but not to limit the scope of other embodiments of this disclosure. It is to be understood that the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. All terms and phrases, including technical and scientific terms and phrases, used here have the same meanings as commonly understood by one of ordinary skill in the art to which the embodiments of this disclosure belong. It will be further understood that terms and phrases, such as those defined in commonly-used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined here. In some cases, the terms and phrases defined here may be interpreted to exclude embodiments of this disclosure.

[0018] Examples of an “electronic device” according to embodiments of this disclosure may include at least one of a smartphone, a tablet personal computer (PC), a mobile phone, a video phone, an e-book reader, a desktop PC, a laptop computer, a netbook computer, a workstation, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, a mobile medical device, a camera, or a wearable device (such as smart glasses, a head-mounted device (HMD), electronic clothes, an electronic bracelet, an electronic necklace, an electronic accessory, an electronic tattoo, a smart mirror, or a smart watch). Other examples of an electronic device include a smart home appliance. Examples of the smart home appliance may include at least one of a television, a digital video disc (DVD) player, an audio player, a refrigerator, an air conditioner, a cleaner, an oven, a microwave oven, a washer, a dryer, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV box (such as SAMSUNG HOMESYNC, APPLETV, or GOOGLE TV), a smart speaker or speaker with an integrated digital assistant (such as SAMSUNG GALAXY HOME, APPLE HOMEPOD, or AMAZON ECHO), a gaming console (such as an XBOX, PLAYSTATION, or NINTENDO), an electronic dictionary, an electronic key, a camcorder, or an electronic picture frame. Still other examples of an electronic device include at least one of various medical devices (such as diverse portable medical measuring devices (like a blood sugar measuring device, a heartbeat measuring device, or a body temperature measuring device), a magnetic resource angiography (MRA) device, a magnetic resource imaging (MRI) device, a computed tomography (CT) device, an imaging device, or an ultrasonic device), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), an automotive infotainment device, a sailing electronic device (such as a sailing navigation device or a gyro compass), avionics, security devices, vehicular head units, industrial or home robots, automatic teller machines (ATMs), point of sales (POS) devices, or Internet of Things (IoT) devices (such as a bulb, various sensors, electric or gas meter, sprinkler, fire alarm, thermostat, street light, toaster, fitness equipment, hot water tank, heater, or boiler). Other examples of an electronic device include at least one part of a piece of furniture or building/structure, an electronic board, an electronic signature receiving device, a projector, or various measurement devices (such as devices for measuring water, electricity, gas, or electromagnetic waves). Note that, according to various embodiments of this disclosure, an electronic device may be one or a combination of the above-listed devices. According to some embodiments of this disclosure, the electronic device may be a flexible electronic device. The electronic device disclosed here is not limited to the above-listed devices and may include new electronic devices depending on the development of technology.

[0019] In the following description, electronic devices are described with reference to the accompanying drawings, according to various embodiments of this disclosure. As used here, the term “user” may denote a human or another device (such as an artificial intelligent electronic device) using the electronic device.

[0020] Definitions for other certain words and phrases may be provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

[0021] None of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claim scope. The scope of patented subject matter is defined only by the claims. Moreover, none of the claims is intended to invoke 35 U.S.C. § 112(f) unless the exact words “means for” are followed by a participle. Use of any other term, including without limitation “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller,” within a claim is understood by the Applicant to refer to structures known to those skilled in the relevant art and is not intended to invoke 35 U.S.C. § 112(f).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] For a more complete understanding of this disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

[0023] FIG. 1 illustrates an example network configuration including an electronic device in accordance with this disclosure;

[0024] FIG. 2 illustrates an example generative video summarization system architecture in accordance with this disclosure;

[0025] FIG. 3 illustrates the multimodal scene detection operation in accordance with this disclosure;

[0026] FIG. 4 illustrates an example multimodal semantic alignment and analysis process in accordance with this disclosure;

[0027] FIG. 5 illustrates an example multimodal video generation process in accordance with this disclosure;

[0028] FIG. 6 illustrates an example generative scene representation creation process in accordance with this disclosure;

[0029] FIG. 7 illustrates an example generative scene representation model process in accordance with this disclosure;

[0030] FIG. 8 illustrates an example method for generative multimedia creation in accordance with this disclosure;

[0031] FIG. 9 illustrates an example method for genre-specific multimodal generative multimedia creation in accordance with this disclosure; and

[0032] FIG. 10 illustrates an example method for generative original multimedia creation in accordance with this disclosure.

DETAILED DESCRIPTION

[0033] FIGS. 1 through 10, discussed below, and the various embodiments of this disclosure are described with reference to the accompanying drawings. However, it should be appreciated that this disclosure is not limited to these embodiments, and all changes and/or equivalents or replacements thereto also belong to the scope of this disclosure. The same or similar reference denotations may be used to refer to the same or similar elements throughout the specification and the drawings.

[0034] As noted above, streaming television has become popular for users to consume linear television video content. However, users increasingly have less time to watch long-form video content, leading to the rise in short-form video content and services. There is, therefore, a need to provide users with a way to consume television content in a short-form format.

[0035] This disclosure provides for personalized generative video summarization systems and methods that utilize the rich video content on television streaming platforms to create short-form content for display on an electronic device. Various embodiments of this disclosure include creating a short-form summary video based on a long-form television (or other video content) program using a genre-specific multimodal transformer (GMT) model and a streaming multimodal video generation model. In some embodiments, the summary video can be used in a current program guide system of the television

streaming system to provide a more intuitive guide for users when they try to select a program to watch. That is, the summary video can take the place of, or be displayed along with, a TV guide interface. Various embodiments of this disclosure also allow for creating short-form videos based on user preferences. For example, even without watching the long-form original video content, a user may wish to watch just short-form videos that are generated specially for user by analyzing the user's historical TV watching behavior.

[0036] The generative video summarization system of this disclosure thus provides for the creation of user specific summarizations of original television programs to provide an enhanced user experience when searching the television content, and can also enhance the content of the generated video based on user preferences to make the summary video more attractive to a specific user. The various embodiments of this disclosure allow for splitting of original long video into different scenes according to user preference based semantic analysis, generation of a summary video without editing the original video, and maintaining semantic and temporal consistence in the generated summary video. This is accomplished by using the GMT model that is configured to execute scene selection and genre-based semantic analysis at the same time and by using the streaming multimodal video generation model that is configured to create a generative artificial intelligence video according to the original input video at any length.

[0037] In various embodiments, the GMT model is configured to perform multimodal position encoding that allows the GMT model to run scene level semantic understanding at the content and temporal levels. In various embodiments, the GMT model is configured to perform self-attention with multimodal alignment that allows the GMT model to execute a comprehensive and cross-domain semantic analysis. In various embodiments, the GMT model is trained using a loss function with a classification loss and a multi-label classification loss that enables the GMT model to run multi-task operations during end-to-end training simultaneously. In various embodiments, a multimodal alignment loss is used to allow the GMT model to improve consistency among different inputs during training.

[0038] In various embodiments, the streaming multimodal video generation model includes a transformer-based multimodal stream control model configured to generate prompt text according to frame captioning and closed captioning at time t in the original video, to provide frame level temporal control. In various embodiments, the streaming multimodal video generation model includes a consistency control model that takes the generated frame from time $t-1$ and the prompt text from time t and utilizes a pre-trained image encoder and pre-trained text encoder to calculate the contrastive loss to control a diffusion model while the diffusion model generates a new frame for the summary video.

[0039] Note that while some of the embodiments discussed below are described in the context of use in consumer electronic devices (such as smartphones), this is merely one example. It will be understood that the principles of this disclosure may be implemented in any number of other suitable contexts and may use any suitable device or devices. Also note that while some of the embodiments discussed below are described based on the assumption that one device (such as a server) performs training of a machine learning model that is deployed to one or more other devices (such as one or more consumer electronic devices), this is also merely one example. It will be understood that the principles of this disclosure may be implemented using any number of devices, including a single device that both trains and uses a machine learning model. In general, this disclosure is not limited to use with any specific type(s) of device(s).

[0040] FIG. 1 illustrates an example network configuration **100** including an electronic device in accordance with this disclosure. The embodiment of the network configuration **100** shown in FIG. 1 is for illustration only. Other embodiments of the network configuration **100** could be used without departing from the scope of this disclosure.

[0041] According to embodiments of this disclosure, an electronic device **101** is included in the network configuration **100**. The electronic device **101** can include at least one of a bus **110**, a processor **120**, a memory **130**, an input/output (I/O) interface **150**, a display **160**, a communication interface **170**, or a sensor **180**. In some embodiments, the electronic device **101** may exclude at least one of these components or may add at least one other component. The bus **110** includes a circuit for connecting the components **120-180** with one another and for transferring communications (such as control messages and/or data) between the components.

[0042] The processor **120** includes one or more processing devices, such as one or more microprocessors, microcontrollers, digital signal processors (DSPs), application specific integrated circuits (ASICs), or field programmable gate arrays (FPGAs). In some embodiments, the processor **120** includes one or more of a central processing unit (CPU), an application processor (AP), a communication processor (CP), or a graphics processor unit (GPU). The processor **120** is able to perform control on at least one of the other components of the electronic device **101** and/or perform an operation or data processing relating to communication or other functions. As described in more detail below, the processor **120** may perform various operations related to personalized generative video summarization.

[0043] The memory **130** can include a volatile and/or non-volatile memory. For example, the memory **130** can store commands or data related to at least one other component of the electronic device **101**. According to embodiments of this disclosure, the memory **130** can store software and/or a program **140**. The program **140** includes, for example, a kernel **141**, middleware **143**, an application programming interface (API) **145**, and/or an application program (or “application”) **147**. At least a portion of the kernel **141**, middleware **143**, or API **145** may be denoted an operating system (OS).

[0044] The kernel **141** can control or manage system resources (such as the bus **110**, processor **120**, or memory **130**) used to perform operations or functions implemented in other programs (such as the middleware **143**, API **145**, or application **147**). The kernel **141** provides an interface that allows the middleware **143**, the API **145**, or the application **147** to access the individual components of the electronic device **101** to control or manage the system resources. The application **147** may support various functions related to personalized generative video summarization. These functions can be performed by a single application or by multiple applications that each carries out one or more of these functions. The middleware **143** can function as a relay to allow the API **145** or the application **147** to communicate data with the kernel **141**, for instance. A plurality of applications **147** can be provided. The middleware **143** is able to control work requests received from the applications **147**, such as by allocating the priority of using the system resources of the electronic device **101** (like the bus **110**, the processor **120**, or the memory **130**) to at least one of the plurality of applications **147**. The API **145** is an interface allowing the application **147** to control functions provided from the kernel **141** or the middleware **143**. For example, the API **145** includes at least one interface or function (such as a command) for filing control, window control, image processing, or text control.

[0045] The I/O interface **150** serves as an interface that can, for example, transfer commands or data input from a user or other external devices to other component(s) of the electronic device **101**. The I/O interface **150** can also output commands or data received from other component(s) of the electronic device **101** to the user or the other external device.

[0046] The display **160** includes, for example, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a quantum-dot light emitting diode (QLED) display, a microelectromechanical systems (MEMS) display, or an electronic paper display. The display **160** can also be a depth-aware display, such as a multi-focal display. The display **160** is able to display, for example, various contents (such as text, images, videos, icons, or symbols) to the user. The display **160** can include a touchscreen and may receive, for example, a touch, gesture, proximity, or hovering input using an electronic pen or a body portion of the user.

[0047] The communication interface **170**, for example, is able to set up communication between the electronic device **101** and an external electronic device (such as a first electronic device **102**, a second electronic device **104**, or a server **106**). For example, the communication interface **170** can be connected with a network **162** or **164** through wireless or wired communication to communicate with the external electronic device. The communication interface **170** can be a wired or wireless transceiver or any other component for transmitting and receiving signals.

[0048] The wireless communication is able to use at least one of, for example, WiFi, long term evolution (LTE), long term evolution-advanced (LTE-A), 5th generation wireless system (5G), millimeter-wave or 60 GHz wireless communication, Wireless USB, code division multiple access (CDMA), wideband code division multiple access (WCDMA), universal mobile telecommunication system (UMTS), wireless broadband (WiBro), or global system for mobile communication (GSM), as a communication protocol. The wired connection can include, for example, at least one of a universal serial bus (USB), high definition multimedia interface (HDMI), recommended standard 232 (RS-232), or plain old telephone

service (POTS). The network **162** or **164** includes at least one communication network, such as a computer network (like a local area network (LAN) or wide area network (WAN)), Internet, or a telephone network.

[0049] The electronic device **101** further includes one or more sensors **180** that can meter a physical quantity or detect an activation state of the electronic device **101** and convert metered or detected information into an electrical signal. For example, one or more sensors **180** can include one or more cameras or other imaging sensors for capturing images of scenes. The sensor(s) **180** can also include one or more buttons for touch input, one or more microphones, a gesture sensor, a gyroscope or gyro sensor, an air pressure sensor, a magnetic sensor or magnetometer, an acceleration sensor or accelerometer, a grip sensor, a proximity sensor, a color sensor (such as an RGB sensor), a bio-physical sensor, a temperature sensor, a humidity sensor, an illumination sensor, an ultraviolet (UV) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an ultrasound sensor, an iris sensor, or a fingerprint sensor. The sensor(s) **180** can further include an inertial measurement unit, which can include one or more accelerometers, gyroscopes, and other components. In addition, the sensor(s) **180** can include a control circuit for controlling at least one of the sensors included here. Any of these sensor(s) **180** can be located within the electronic device **101**.

[0050] In some embodiments, the first external electronic device **102** or the second external electronic device **104** can be a wearable device or an electronic device-mountable wearable device (such as an HMD). When the electronic device **101** is mounted in the electronic device **102** (such as the HMD), the electronic device **101** can communicate with the electronic device **102** through the communication interface **170**. The electronic device **101** can be directly connected with the electronic device **102** to communicate with the electronic device **102** without involving with a separate network. The electronic device **101** can also be an augmented reality wearable device, such as eyeglasses, that include one or more imaging sensors.

[0051] The first and second external electronic devices **102** and **104** and the server **106** each can be a device of the same or a different type from the electronic device **101**. According to certain embodiments of this disclosure, the server **106** includes a group of one or more servers. Also, according to certain embodiments of this disclosure, all or some of the operations executed on the electronic device **101** can be executed on another or multiple other electronic devices (such as the electronic devices **102** and **104** or server **106**). Further, according to certain embodiments of this disclosure, when the electronic device **101** should perform some function or service automatically or at a request, the electronic device **101**, instead of executing the function or service on its own or additionally, can request another device (such as electronic devices **102** and **104** or server **106**) to perform at least some functions associated therewith. The other electronic device (such as electronic devices **102** and **104** or server **106**) is able to execute the requested functions or additional functions and transfer a result of the execution to the electronic device **101**. The electronic device **101** can provide a requested function or service by processing the received result as it is or additionally. To that end, a cloud computing, distributed computing, or client-server computing technique may be used, for example. While FIG. **1** shows that the electronic device **101** includes the communication interface **170** to communicate with the external electronic device **104** or server **106** via the network **162** or **164**, the electronic device **101** may be independently operated without a separate communication function according to some embodiments of this disclosure.

[0052] The server **106** can include the same or similar components **110-180** as the electronic device **101** (or a suitable subset thereof). The server **106** can support to drive the electronic device **101** by performing at least one of operations (or functions) implemented on the electronic device **101**. For example, the server **106** can include a processing module or processor that may support the processor **120** implemented in the electronic device **101**. As described in more detail below, the server **106** may perform various operations related to personalized generative video summarization.

[0053] Although FIG. **1** illustrates one example of a network configuration **100** including an electronic device **101**, various changes may be made to FIG. **1**. For example, the network configuration **100** could include any number of each component in any suitable arrangement. In general, computing and communication systems come in a wide variety of configurations, and FIG. **1** does not limit the scope of this disclosure to any particular configuration. Also, while FIG. **1** illustrates one operational environment in which various features disclosed in this patent document can be used, these features could be used in

any other suitable system.

[0054] FIG. 2 illustrates an example generative video summarization system architecture **200** in accordance with this disclosure. For ease of explanation, the architecture **200** shown in FIG. 2 is described as being implemented on or supported by the electronic device **101** in the network configuration **100** of FIG. 1. However, the architecture **200** shown in FIG. 2 could be used with any other suitable device(s) and in any other suitable system(s), such as when the architecture **200** is implemented on or supported by the server **106**.

[0055] As shown in FIG. 2, the architecture **200** includes a store of multimedia content **202**, including multimedia content items such as one or more television programs. The architecture includes a video backbone **204** and an audio backbone **206**. The video of a multimedia content item is processed by the video backbone **204** and the audio of the multimedia content item is processed by the audio backbone **206**. In various embodiments, the audio backbone **206** can include an automated speech recognition (ASR) model that converts acoustic information in the audio to text.

[0056] The architecture **200** also includes a multimodal scene detection operation **208** that is performed by taking both video data from the video backbone **204** and audio data (which can be text data) from the audio backbone **206** and cuts or divides the original video into scenes. For example, FIG. 3 illustrates the multimodal scene detection operation **208** in accordance with this disclosure. For ease of explanation, the multimodal scene detection operation **208** shown in FIG. 3 is described as being implemented on or supported by the electronic device **101** in the network configuration **100** of FIG. 1. However, the multimodal scene detection operation **208** shown in FIG. 3 could be used with any other suitable device(s) and in any other suitable system(s), such as when the multimodal scene detection operation **208** is implemented on or supported by the server **106**.

[0057] As shown in FIG. 3, the multimodal scene detection operation **208** first runs scene detection separately on the video data from the video backbone **204** and the audio data from the audio backbone **206**. For instance, as shown in FIG. 3, video scene detection executed on the video data from the video backbone **204** results in the creation of a plurality of video scenes **302**, and audio scene detection separately executed on the audio data from the audio backbone **206** results in the creation of a plurality of sentences **304**. Scene alignment is then executed by the multimodal scene detection operation **208** to combine the two scene detection results **302**, **304** into multimodal scenes **306** that include both video and audio data. The multimodal scene detection operation **208** helps to prevent flickering issues between the video and audio, such as a video scene transitioning to another video scene before a sentence from the audio data is completed, which can result in the sentence being cut off unnaturally and before it is fully presented to the user.

[0058] Referring again to FIG. 2, the architecture **200** includes a genre-specific multimodal transformer (GMT) model **210**. As described in this disclosure, outputs from the multimodal scene detection operation **208**, such as at least a portion of the combined multimodal scenes, are provided to the GMT model **210** and the GMT model **210** performs scene selection and genre-based semantic analysis simultaneously to perform multimodal position encoding to map video, audio, and genre information to a same latent space, and perform self-attention with multimodal alignment to merge information from different modalities.

[0059] The architecture **200** also includes a scene selection model **212** that receives outputs from the GMT model **210**, such as outputs including scene importance scores and semantic analysis results, and selects scenes for a specific user according to this user's personal preferences. As described in this disclosure, the architecture **200** further includes a streaming multimodal video generation model **214** that is used to generate a summary video **216** based on the scenes selected by the scene selection model **212**. The streaming multimodal video generation model **214** provides temporal and semantic consistence during video generation.

[0060] The architecture **200** can achieve genre-specific video summarization gains across multiple genres compared to other architectures. For example, even with a varying number of genres and varying video length, the GMT model **210** can outperform other baseline architectures and can provide consistent results even when the number of genres increases. Because the GMT model **210** can utilize multimodal scene detection information which considers both scene and sentence completeness in the scene segmentation, the GMT model **210** provides improved multimodal understanding and an improved user experience.

[0061] Although FIG. 2 illustrates one example of a generative video summarization system architecture **200**, various changes may be made to FIG. 2. For example, various components and functions in FIG. 2 may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired.

[0062] Although FIG. 3 illustrates one example of a multimodal scene detection operation **208**, various changes may be made to FIG. 3. For example, various components and functions in FIG. 3 may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired.

[0063] FIG. 4 illustrates an example multimodal semantic alignment and analysis process **400** in accordance with this disclosure. For case of explanation, the process **400** is described as involving the use of the electronic device **101** in the network configuration **100** of FIG. 1. However, the process **400** may be used with any other suitable electronic device (such as the server **106**) or a combination of devices (such as the electronic device **101** and the server **106**) and in any other suitable system(s).

[0064] As shown in FIG. 4, the multimodal semantic alignment and analysis process **400** includes that utilizes a multimodal semantic automatic alignment network and the GMT model **210** that takes as inputs both the video and ASR text features as well as genre features. As shown in FIG. 4, the video, audio text, and genre inputs each have their feature or token embedding **402**, sequence position embedding **404**, and multimodal segment embedding **406** to map them into the same latency space. This multimodal position embedding is used to keep the completeness of both video and audio to prevent flicker issues in the final summary video output.

[0065] As also shown in FIG. 4, the token embedding **402** includes tokens for the genre information, the video frames, and the audio text sentences. For example, as shown in FIG. 4, the genre information can include classification tokens denoted and starting with a classification tag ([CLS]) following by genre information pertaining to the genre of scenes in the video. The video frames and audio text sentences are also indicated in the token embedding **402**. The scenes, as determined for example using the multimodal scene detection operation **208**, are included in the multimodal segment embedding **406** and are aligned with the different sequence position embeddings **404** and are also aligned with the frames and audio text sentences in the token embeddings **402** that are associated with the particular scene. For instance, as shown in FIG. 4, scene **1** includes the first frame of the video and the first sentence of the audio text, and scene **1** is tagged as being in the “drama” genre. Scene **2** in the example of FIG. 4 includes the second, third, and fourth frames of the video and the second and third sentences of the audio text, and scene **2** is tagged as being in the “Sci-Fi” genre.

[0066] As illustrated in FIG. 4, there can be any number of token embeddings **402**, sequence position embeddings **404**, and multimodal segment embeddings **406**, such as depending on the length of the original input video.

[0067] The GMT model **210** utilizes an alignment-guided self-attention structure that is used to run semantic analysis and scene alignment for the input video to create a genre-guided multimodal scene attention matrix **408** that functions like a map to align the genres, frames, and scenes in the inputs. Multi-task prediction follows the alignment-guided self-attention structure to predict the cut position for scenes in original video and the corresponding genre for each scene. As illustrated in FIG. 4, the dashed line parts of the matrix **408** indicate a “1” and the other parts indicate a “0” for the values in the matrix **408**.

[0068] In various embodiments, because user preferences for systems like automatic content recognition (ACR) systems are typically genre-based, the aim of GMT model **210** is to recognize the most important scenes in a given video by considering the input genres that reflect the user's preferences. As shown in FIG. 4, the GMT model **210** takes in multiple modalities (e.g., video and closed caption sentences) comprising M video frames and N sentences. As described above, multimodal scene detection is initially conducted, and each video frame and sentence are assigned to a specific scene along with its corresponding start and end times. The overall architecture can be structured as three components: input embedding, multimodal alignment and fusion, and a loss function.

[0069] For the model input, given an untrimmed video V, it includes M frames with their corresponding visual features $f_{\text{sub}.i}$ ($i=1, 2, \dots, M$) and N sentences with their text features $s_{\text{sub}.j}$ ($j=1, 2, \dots, N$), along with the user-preferred genre $g_{\text{sub}.l}$ ($l=1, 2, \dots, U$). The multimodal position encoding shown in FIG. 4 is trainable and used during the input phase. Each frame and sentence correspond to one of the K

scenes denoted as $s.\text{sub}.k$ ($k=1, 2, \dots, K$). It will be understood that a scene may encompass multiple frames f and sentences s that share similar semantics (e.g., genre). After incorporating these positional embeddings for frames and sentences, the final input sequences, comprising genres, frames, and sentences, are concatenated, which can be denoted as $X \in \mathbb{R}^{\text{sup}.(U+M+N) \times C}$.

[0070] As also indicated above, the alignment-guided self-attention mechanism to merge information from different modalities is utilized through a masked self-attention operation with a corresponding mask $A \in \mathbb{R}^{\text{sup}.(U+M+N) \times (U+M+N)}$. This modeling within the same modality enabling features to focus on one another, resulting in all entries related to intra-modality attention being assigned a value of 1 in the attention mask. Concerning the cross-modality attention between video and textual inputs, only entries linked to the same multimodal segment are marked with “1”. In contrast to previous methods that establish cross-modality attention based on frame and closed caption co-occurrence, the process **400** determines cross-modality attention based on the multimodal scene detection. The frames and sentences are conditioned on the input genres, the attention is set to 1, and the attention mask is utilized on the attention matrix computed using a self-attention process, which can be represented as follows.

$$[00001] Q = XW_Q, K = XW_K, V = XW_V, D_{i,j} = \frac{A_{i,j} \exp(Q_i K_j^T / \sqrt{D})}{\sum_k A_{i,k} \exp(Q_i K_k^T / \sqrt{D})}, Z = X + DVW_O$$

Here, i and j ranging from 1 to $U+M+N$ represent the indices of entries in the matrix. X refers to the input features that include genre, frames, and sentences. The matrices $W_{\text{sub}.Q}$, $W_{\text{sub}.K}$, $W_{\text{sub}.V}$, and $W_{\text{sub}.O}$ of size $\mathbb{R}^{\text{sup}.C \times C}$ are utilized for projecting queries, keys, values, and generating outputs. This allows for explicitly exploiting the alignment relationships between genre, video and text in multimodal scene detection, resulting in improved alignment between modalities.

[0071] As further shown in FIG. 4, a video projection operation **410** and a text projection operation **412** of the GMT model **210** are used to output the labels for each video frame and audio sentence between 1 (foreground) and 0 (background) based on the genre for the particular video frame or audio sentence. These labels are used to provide an importance score for each scene in the original video content.

[0072] A loss function can be used to train the GMT model **210** that incorporates a classification loss, a multi-label classification loss, and a multimodal alignment loss. For the classification loss, focal loss for the classification of importance scores is employed, which addresses problems of imbalanced classes by decreasing the loss for correctly classified samples. The classification loss can be represented as follows.

$$[00002] \mathcal{L}_{\text{cls}_m} = -\begin{cases} (1-p) \log(p), & \text{if } y = 1 \\ -(1-p) \log(1-p), & \text{if } y = 0 \end{cases} \quad \mathcal{L}_{\text{cls}} = \mathcal{L}_{\text{cls}_{\text{video}}} + \mathcal{L}_{\text{cls}_{\text{text}}}$$

Here, m represents the modality, p represents the predicted score for each frame/sentence and y denotes the true label, which can be 0 or 1 as it denotes foreground/background. The overall classification loss is the cumulative sum of the losses for each modality. The text label is obtained from video summarization labels. The text label $y=1$ signifies that the sentence had a temporal overlap with the final video summarization timeline.

[0073] For the multi-label classification loss, genre prediction is conducted for each visual and textual token. The genre label can be single or multiple in this context. To address this, Binary Cross Entropy (BCE) loss can be employed for prediction. In the case of K labels (genres), the network independently predicts the logits for the i -th label, where $p_{\text{sub}.i}$ represents the predicted score for each frame or sentence. Let $y_{\text{sub}.i}$ indicate the label for the i -th class, such that the multi-label classification loss can be represented as follows.

$$[00003] \mathcal{L}_{\text{mul}_m} = -\frac{1}{K} \begin{cases} -\log(p_i), & \text{if } y_i = 1 \\ -\log(1-p_i), & \text{if } y_i = 0 \end{cases} \quad \mathcal{L}_{\text{mul}} = \mathcal{L}_{\text{mul}_{\text{video}}} + \mathcal{L}_{\text{mul}_{\text{text}}}$$

[0074] The multimodal alignment loss is used to achieve accurate summarization that maintains consistency between visual scenes and complete sentences, where the method guides the prediction process to ensure alignment between visual and textual elements. Like contrastive loss, cosine similarity between tokens in the visual and textual domains is leveraged to bring positive pairs closer together while pushing negative pairs apart. Positive pairs are determined by ground truth aligned visual-text pairs that correspond to the same multimodal scene detection outcome. In the matrix **408** ($M \in \mathbb{R}^{\text{sup}.M \times N}$), $M_{\text{sub}.ij}=1$ if frame $f_{\text{sub}.i}$ and sentence $s_{\text{sub}.j}$ are part of the same multimodal scene. Ultimately, the

model is trained to make more aligned predictions within an untrimmed video V. The multimodal alignment loss function  custom-character.sub.align can be represented as follows.

[00004]

$$\mathcal{L}_{\text{align}} = \frac{1}{M} \sum_{i=1}^M \log \frac{e^{f_i \cdot s_k}}{e^{f_i \cdot s_k} + e^{f_i \cdot s_k^{\text{imp}}}} + \frac{1}{N} \sum_{j=1}^N \log \frac{e^{f_i \cdot s_j}}{e^{f_i \cdot s_j} + e^{f_i^{\text{imp}} \cdot s_j}}$$

Here, f.sub.i.sup.imp and s.sub.k.sup.imp denote imposter pairs from the two modalities, sampled from the same video but not belonging to the same multimodal scene. The multimodal alignment loss function aims to distinguish between genuine positive embedding pairs and imposter negative pairs within each video V. By separating positive and negative samples, the spaces of positive and negative samples are enumerated independently. In one scenario, text samples that are in the same scene are paired with various negative video samples, while in the other, video samples that are in the same scene are paired with various negative text samples. The hyperparameter **8** serves as a margin and is chosen empirically. [0075] The final GMT model **210** is trained to minimize the sum of the classification, multi-label classification, and multimodal alignment losses, which can be represented as follows.

[00005] $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{sub.1}} \mathcal{L}_{\text{mul}} + \lambda_{\text{sub.1}} \mathcal{L}_{\text{align}}$

Here, $\lambda_{\text{sub.1}}$ and $\lambda_{\text{sub.1}}$ are hyperparameters.

[0076] The GMT model **210** outputs scene importance scores S.sub.importance and predicted genres G.sub.p. The scene selection model **212** described with respect to FIG. **2** takes the outputs from the GMT model **210** and uses the outputs to select the scenes to that will be used to create the final summarization video. The scene selection model **212** calculates a final user-specific scene score S using a function that uses the scene importance score summed with a similarity of the predicted genre G.sub.p for the scene output by the GMT model **210** and a known user genre preference G.sub.u, which can be represented as follows.

[00006] $S = S_{\text{importance}} + \text{Similarity}(G_p + G_u)$

Again, here, S.sub.importance is the scene importance score from the GMT model **210**, G.sub.p is the predicted genre from the GMT model **210** and G.sub.u is the user preference (genre-based). A T5 text encoder can be used here to run embeddings for G.sub.p and G.sub.u, and cosine similarity can be used to calculate the similarity score between them.

[0077] When scene score is calculated, the top N scenes are selected for use in creating the final summarization video. N is according to the length of the summarization video (e.g., a parameter can be set such that the summarization video is particular length, such as a parameter indicating the summarization video length should be 15% length of original video). The length can be adjustable in the pipeline.

[0078] Although FIG. **4** illustrates one example of a multimodal semantic alignment and analysis process **400**, various changes may be made to FIG. **4**. For example, various components and functions in FIG. **4** may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired. It will be understood that the number of and types of genres, video frames, and audio sentences shown in FIG. **4** can vary based on the original input video.

[0079] FIG. **5** illustrates an example multimodal video generation process **500** in accordance with this disclosure. For ease of explanation, the process **500** is described as involving the use of the electronic device **101** in the network configuration **100** of FIG. **1**. However, the process **500** may be used with any other suitable electronic device (such as the server **106**) or a combination of devices (such as the electronic device **101** and the server **106**) and in any other suitable system(s).

[0080] As shown in FIG. **5**, the process **500** uses the streaming multimodal video generation model **214**, also described with respect to FIG. **2**, to generate a summary final multimedia content result **501**, e.g., a summarization video. The streaming multimodal video generation model **214** includes a transformer-based multimodal stream control model **502**, a text-to-video model **504** (e.g., a diffusion model), and a consistency control model **506**.

[0081] The transformer-based multimodal stream control model **502** generates prompt text using a transformer encoder **508** and a transformer decoder **510** according to frame captioning **512** and closed captioning **514** at time t of the original input video and using the top N scenes selected for use in creating the final summarization video by the scene selection model **212**. The transformer-based multimodal

stream control model **502** provides frame level temporal control to the streaming multimodal video generation model **214**.

[0082] The text-to-video model **504** receives the prompt text at time t and generates an image frame **515** at time t according to each prompt text received. The consistency control model **506** includes a pretrained image encoder **516** and a pretrained text encoder **518**. The pretrained image encoder **516** takes as input the generated frame from time $t-1$ and the pretrained text encoder **518** takes as input the prompt text from time t . As shown in FIG. 5, the output from the image encoder **516** is processed through an embedding layer **520** and a fully connected network (FCN) layer **522**. Likewise, the output from the text encoder **518** is processed through an embedding layer **524** and an FCN layer **526**. The consistency control model **506** utilizes the outputs from the FCNs **522**, **526** to calculate an image-text contrastive loss **528**. As shown in FIG. 5, the image-text contrastive loss **528** is used to control the text-to-video model **504** while it is generating a next new frame. A frame post-processing operation **530** is then performed on the generated frames **515** to construct the final video **501**. The resulting final video **501** thus includes a series of completely AI-generated images, i.e., although the final video **501** is based on the original input video, the images are not from the original input video, but are AI-generated based on the prompt text.

[0083] Although FIG. 5 illustrates one example of a multimodal video generation process **500**, various changes may be made to FIG. 5. For example, various components and functions in FIG. 5 may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired.

[0084] In some embodiments of this disclosure, using a variation of the streaming multimodal video generation model **214**, it can also be possible to not just create user specific generative summary videos for television programs, but to also create original content. FIG. 6 illustrates an example generative scene representation creation process **600** in accordance with this disclosure. For case of explanation, the process **600** is described as involving the use of the electronic device **101** in the network configuration **100** of FIG. 1. However, the process **600** may be used with any other suitable electronic device (such as the server **106**) or a combination of devices (such as the electronic device **101** and the server **106**) and in any other suitable system(s).

[0085] The process **600** includes tracking user behavior **602**, such as user TV-watching behavior over a particular time period. A behavior description model **604** summarizes the user behavior **602**, such as constructing a summary description in one or more paragraphs. A script generation model **606** then transforms the summary description into script, e.g., a movie script. A generative scene representation model **614** uses the script as prompt text to generate a video **608**, such as a short movie based on the movie script, to provide for display a user's own preference-based video. In this way, original content can be created based on the user's preferences and without needing to use an original video input. For example, if the user's behavior includes watching both comedies and science fiction shows on television, a movie script can be created that includes a character from a science fiction show the user watches performing comedic acts.

[0086] Although FIG. 6 illustrates one example of a generative scene representation creation process **600**, various changes may be made to FIG. 6. For example, various components and functions in FIG. 6 may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired.

[0087] FIG. 7 illustrates an example generative scene representation model process **700** in accordance with this disclosure. For case of explanation, the process **700** is described as involving the use of the electronic device **101** in the network configuration **100** of FIG. 1. However, the process **700** may be used with any other suitable electronic device (such as the server **106**) or a combination of devices (such as the electronic device **101** and the server **106**) and in any other suitable system(s).

[0088] As shown in FIG. 7, the process **700** uses the generative scene representation model **614** described in FIG. 6. As also shown in FIG. 7, the generative scene representation model **614** can be similar to the streaming multimodal video generation model **214**, except that, instead of generating prompt text using the stream control model **502**, the script generation model **606** is used to create prompt text for the generative scene representation model **614**. The generative scene representation model **614** thus includes a text-to-video model **704** (e.g., a diffusion model), and a consistency control model **706**.

[0089] As shown in FIG. 7, the process **700** includes the script generation model **606** providing prompt

text at time t to the generative scene representation model **614**. The text-to-video model **704** receives the prompt text at time t and generates an image frame **715** at each time t according to its prompt text/script at that time. The consistency control model **706** includes a pretrained image encoder **716** and a pretrained text encoder **718**. The pretrained image encoder **716** takes as input the generated frame from time $t-1$ and the pretrained text encoder **718** takes as input the prompt text from time t . As shown in FIG. 7, the output from the image encoder **716** is processed through an embedding layer **720** and an FCN layer **722**. Likewise, the output from the text encoder **718** is processed through an embedding layer **724** and an FCN layer **726**.

[0090] The consistency control model **706** utilizes the outputs from the FCNs **722**, **726** to calculate an image-text contrastive loss **728**. As shown in FIG. 7, the image-text contrastive loss **728** is used to control the text-to-video model **704** while it is generating a next new frame. After all frames have been generated according to script generation, a frame post-processing operation **730** is performed on the generated frames **717** to combine them together to construct the final video **701**. The resulting final video **701** thus includes a series of completely AI-generated images, generated based on the script used as the prompt text. The process **700** thus allows for the generation of video content (such as short movies) based on just user behavior, and without relying on videos from other sources.

[0091] Although FIG. 7 illustrates one example of a generative scene representation model process **700**, various changes may be made to FIG. 7. For example, various components and functions in FIG. 7 may be combined, further subdivided, replicated, or rearranged according to particular needs. Also, one or more additional components and functions may be included if needed or desired.

[0092] FIG. 8 illustrates an example method **800** for generative multimedia creation in accordance with this disclosure. For ease of explanation, the method **800** shown in FIG. 8 is described as being performed using the electronic device **101** in the network configuration **100** of FIG. 1. However, the method **800** could be performed using any other suitable device(s), such as the server **106**, and in any other suitable system(s).

[0093] At step **802**, prompt text is created based on one or more user preferences. In various embodiments, this can include the processor **120** of the electronic device **101** executing the stream control model **502** of the streaming multimodal video generation model **214** to generate prompt text based on one or more outputs from the GMT model **210**. In some embodiments, step **802** can include the processor **120** executing using the script generation model **606** to generate the prompt text.

[0094] At step **804**, image frames are generated based on the prompt text. This can include the processor executing the text-to-video model **504** or **704** and the consistency control model **506** or **706** to generate the image frames. At step **806**, the image frames are compiled into a multimedia content result, such as a summarization video or original video content. This can include the processor **120** executing the frame post-processing operation **530** or **730**. At step **808**, the multimedia content result can be displayed on the electronic device **101** or some other electronic device.

[0095] Although FIG. 8 illustrates one example of a method **800** for generative multimedia creation, various changes may be made to FIG. 8. For example, while shown as a series of steps, various steps in FIG. 8 could overlap, occur in parallel, occur in a different order, or occur any number of times (including zero times).

[0096] FIG. 9 illustrates an example method **900** for genre-specific multimodal generative multimedia creation in accordance with this disclosure. For case of explanation, the method **900** shown in FIG. 9 is described as being performed using the electronic device **101** in the network configuration **100** of FIG. 1. However, the method **900** could be performed using any other suitable device(s), such as the server **106**, and in any other suitable system(s).

[0097] At step **902**, video data and audio data associated with a multimedia content item are acquired, such as video data and audio data associated with a television program. At step **904**, a first scene detection process is performed on the video data to produce first scene detection results, and a second scene detection process is performed on the audio data to produce second scene detection results. At step **906**, combined scene detection results are determined based on the first scene detection results and the second scene detection results. Steps **904** and **906** can include the processor **120** of the electronic device **101** executing the multimodal scene detection operation **208**.

[0098] At step **908**, at least a portion of the combined scene detection results is input into a genre-specific

multimodal transformer (GMT) model, such as the GMT model **210**. This can include the processor **120** executing the GMT model **210** as described in this disclosure, such as to use the GMT model **210** to perform an alignment of the first scene detection results, the second scene detection results, and genre data, including mapping the first scene detection results, the second scene detection results, and the genre data into a same latent space, to perform a self-attention process to merge the first scene detection results, the second scene detection results, and the genre data, and to output genre predictions and importance scores.

[0099] At step **910**, one or more scene importance metrics are determined based on the one or more user preferences and the one or more outputs from the GMT model. As described in this disclosure, the one or more outputs from the GMT model can include a scene importance score and a predicted genre. Step **910** can include the processor **120** executing the scene selection model **212**.

[0100] At step **912**, a transformer-based model, such as the stream control model **502**, is used to create prompt text using inputs based on the one or more outputs from the GMT model. At step **914**, a diffusion model is used to generate at least one image frame based on the prompt text. At step **916**, a consistency control model is used to evaluate a contrastive loss using the prompt text and a previously generated image frame from the diffusion model. It will be understood that steps **912-916** could be performed any number of times to create any number of video frames.

[0101] At step **918**, post-processing is performed on the image frames to compile the image frames into a video. This can include the processor **120** executing the frame post-processing operation **530**. At step **920**, a multimedia content result is generated from the post-processed image frames as a summarization of the multimedia content item.

[0102] Although FIG. **9** illustrates one example of a method **900** for genre-specific multimodal generative multimedia creation, various changes may be made to FIG. **9**. For example, while shown as a series of steps, various steps in FIG. **9** could overlap, occur in parallel, occur in a different order, or occur any number of times (including zero times).

[0103] FIG. **10** illustrates an example method **1000** for generative original multimedia creation in accordance with this disclosure. For case of explanation, the method **1000** shown in FIG. **10** is described as being performed using the electronic device **101** in the network configuration **100** of FIG. **1**. However, the method **1000** could be performed using any other suitable device(s), such as the server **106**, and in any other suitable system(s).

[0104] At step **1002**, one or more user preferences are tracked over a period of time, such as television program watching habits of a user like genre of content watched. At step **1004**, a user behavior description is generated using a behavior description model, such as the behavior description model **604**, summarizing the tracked one or more user preferences over the period of time. This can include the processor **120** of the electronic device **101** executing the behavior description model **604**.

[0105] At step **1006**, a script using the user behavior description is generated where the script is to be used as prompt text for a streaming multimodal video generation model. This can include the processor **120** executing the script generation model **606** and providing the script to the generative scene representation model **614**. At step **1008**, a diffusion model, such as the text-to-video model **704** of the generative scene representation model **614**, is used to receive the script as prompt text and generate at least one image frame based on the prompt text. This can include the processor **120** executing the text-to-video model **704** of the generative scene representation model **614** to generate the image frame.

[0106] At step **1010**, a consistency control model, such as the consistency control model **706**, is used to evaluate a contrastive loss using the prompt text and a previously generated image frame from the diffusion model. It will be understood that steps **1008** and **1010** could be performed any number of times to create any number of video frames.

[0107] At step **1012**, frame post-processing is performing on a plurality of image frames output by the multimodal video generation model and a multimedia content result is output. This can include the processor **120** executing the frame post-processing operation **730** and combining the image frames into the multimedia content result.

[0108] Although FIG. **10** illustrates one example of a method **1000** for generative original multimedia creation, various changes may be made to FIG. **10**. For example, while shown as a series of steps, various steps in FIG. **10** could overlap, occur in parallel, occur in a different order, or occur any number of times

(including zero times).

[0109] It should be noted that the functions shown in FIGS. 2 through 10 or described above can be implemented in an electronic device 101, 102, 104, server 106, or other device(s) in any suitable manner. For example, in some embodiments, at least some of the functions shown in FIGS. 2 through 10 or described above can be implemented or supported using one or more software applications or other software instructions that are executed by the processor 120 of the electronic device 101, 102, 104, server 106, or other device(s). In other embodiments, at least some of the functions shown in FIGS. 2 through 10 or described above can be implemented or supported using dedicated hardware components. In general, the functions shown in FIGS. 2 through 10 or described above can be performed using any suitable hardware or any suitable combination of hardware and software/firmware instructions. Also, the functions shown in FIGS. 2 through 10 or described above can be performed by a single device or by multiple devices.

[0110] Although this disclosure has been described with reference to various example embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that this disclosure encompass such changes and modifications as fall within the scope of the appended claims.

Claims

1. A method comprising: generating a multimedia content result using a streaming multimodal video generation model, including: creating prompt text based on one or more user preferences; generating image frames based on the prompt text; compiling the image frames into the multimedia content result; and displaying the multimedia content result on a display.
2. The method of claim 1, further comprising: acquiring video data and audio data associated with a multimedia content item; performing a first scene detection process on the video data to produce first scene detection results; performing a second scene detection process on the audio data to produce second scene detection results; determining combined scene detection results based on the first scene detection results and the second scene detection results; inputting at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model; and generating the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.
3. The method of claim 2, wherein generating the multimedia content result as the summarization for the multimedia content item includes: determining one or more scene importance metrics based on the one or more user preferences and the one or more outputs from the GMT model, wherein the one or more outputs of the GMT model include a scene importance score and a predicted genre; and generating the summarization based on the one or more scene importance metrics.
4. The method of claim 3, further comprising: performing an alignment of the first scene detection results, the second scene detection results, and genre data, including mapping the first scene detection results, the second scene detection results, and the genre data into a same latent space; and performing a self-attention process to merge the first scene detection results, the second scene detection results, and the genre data.
5. The method of claim 4, wherein the streaming multimodal video generation model includes: a transformer-based model for creating the prompt text using inputs based on the one or more outputs from the GMT model; a diffusion model for generating an image frame based on the prompt text; and a consistency control model for evaluating a contrastive loss using the prompt text and a previously generated image frame from the diffusion model.
6. The method of claim 1, wherein creating the prompt text includes: tracking the one or more user preferences over a period of time; generating, using a behavior description model, a user behavior description summarizing the tracked one or more user preferences over the period of time; and generating a script using the user behavior description.
7. The method of claim 6, wherein the streaming multimodal video generation model includes: a diffusion model for receiving the script as the prompt text and generating an image frame based on the prompt text; a consistency control model for evaluating a contrastive loss using the prompt text and a previously generated image frame from the diffusion model; and a frame post-processing model for combining the

image frames into the multimedia content result.

8. An electronic device comprising: at least one processing device configured to: generate a multimedia content result using a streaming multimodal video generation model, wherein the at least one processing device is further configured to: create prompt text based on one or more user preferences; generate image frames based on the prompt text; compile the image frames into the multimedia content result; and display the multimedia content result on a display.

9. The electronic device of claim 8, wherein the at least one processing device is further configured to: acquire video data and audio data associated with a multimedia content item; perform a first scene detection process on the video data to produce first scene detection results; perform a second scene detection process on the audio data to produce second scene detection results; determine combined scene detection results based on the first scene detection results and the second scene detection results; input at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model; and generate the multimedia content result as a summarization for the multimedia content item using one or more outputs from the GMT model.

10. The electronic device of claim 9, wherein, to generate the multimedia content result as the summarization for the multimedia content item, the at least one processing device is further configured to: determine one or more scene importance metrics based on the one or more user preferences and the one or more outputs from the GMT model, wherein the one or more outputs of the GMT model include a scene importance score and a predicted genre; and generate the summarization based on the one or more scene importance metrics.

11. The electronic device of claim 10, wherein the at least one processing device is further configured to: perform an alignment of the first scene detection results, the second scene detection results, and genre data, including mapping the first scene detection results, the second scene detection results, and the genre data into a same latent space; and perform a self-attention process to merge the first scene detection results, the second scene detection results, and the genre data.

12. The electronic device of claim 11, wherein the streaming multimodal video generation model includes: a transformer-based model configured to create the prompt text using inputs based on the one or more outputs from the GMT model; a diffusion model configured to generate an image frame based on the prompt text; and a consistency control model configured to evaluate a contrastive loss using the prompt text and a previously generated image frame from the diffusion model.

13. The electronic device of claim 8, wherein, to create the prompt text, the at least one processing device is further configured to: track the one or more user preferences over a period of time; generate, using a behavior description model, a user behavior description summarizing the tracked one or more user preferences over the period of time; and generate a script using the user behavior description.

14. The electronic device of claim 13, wherein the streaming multimodal video generation model includes: a diffusion model configured to receive the script as the prompt text and generate an image frame based on the prompt text; a consistency control model configured to evaluate a contrastive loss using the prompt text and a previously generated image frame from the diffusion model; and a frame post-processing model for combining the image frames into the multimedia content result.

15. A non-transitory machine readable medium comprising instructions that when executed cause at least one processor of an electronic device to: generate a multimedia content result using a streaming multimodal video generation model, wherein the instructions when executed further cause the at least one processing device to: create prompt text based on one or more user preferences; generate image frames based on the prompt text; compile the image frames into the multimedia content result; and display the multimedia content result on a display.

16. The non-transitory machine readable medium of claim 15, wherein the instructions when executed further cause the at least one processor of the electronic device to: acquire video data and audio data associated with a multimedia content item; perform a first scene detection process on the video data to produce first scene detection results; perform a second scene detection process on the audio data to produce second scene detection results; determine combined scene detection results based on the first scene detection results and the second scene detection results; input at least a portion of the combined scene detection results into a genre-specific multimodal transformer (GMT) model; and generate the multimedia content result as a summarization for the multimedia content item using one or more outputs

from the GMT model.

17. The non-transitory machine readable medium of claim 16, wherein the instructions that when executed cause the at least one processor of the electronic device to generate the multimedia content result as the summarization for the multimedia content item further include instructions that when executed cause the at least one processor of the electronic device to: determine one or more scene importance metrics based on the one or more user preferences and the one or more outputs from the GMT model, wherein the one or more outputs of the GMT model include a scene importance score and a predicted genre; and generate the summarization based on the one or more scene importance metrics.

18. The non-transitory machine readable medium of claim 17, wherein the instructions when executed further cause the at least one processor of the electronic device to: perform an alignment of the first scene detection results, the second scene detection results, and genre data, including mapping the first scene detection results, the second scene detection results, and the genre data into a same latent space; and perform a self-attention process to merge the first scene detection results, the second scene detection results, and the genre data.

19. The non-transitory machine readable medium of claim 18, wherein the streaming multimodal video generation model includes: a transformer-based model configured to create the prompt text using inputs based on the one or more outputs from the GMT model; a diffusion model configured to generate an image frame based on the prompt text; and a consistency control model configured to evaluate a contrastive loss using the prompt text and a previously generated image frame from the diffusion model.

20. The non-transitory machine readable medium of claim 15, wherein the instructions that when executed cause the at least one processor of the electronic device to create the prompt text further include instructions that when executed cause the at least one processor of the electronic device to: track the one or more user preferences over a period of time; generate, using a behavior description model, a user behavior description summarizing the tracked one or more user preferences over the period of time; and generate a script using the user behavior description.
